Worksheet 2

Measures of central tendency – DETAILED SOLUTIONS

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Detailed solution – Exercise 1

$$\bar{x} = \frac{2(10) + 3(20) + 5(30)}{10} = 23.$$

For growth:

$$1 + g = \sqrt[3]{1.10 \times 0.95 \times 1.20} = 1.078365,$$

so $g \approx \boxed{7.84\%}$. For equal distances:

$$H = \frac{2}{1/60 + 1/90} = \boxed{72 \text{ km/h}}.$$

Detailed solution – Exercise 2

The count densities are $1, 3, 4, 8, 12, 7, 5, 1, 1$, so the modal class is $\boxed{[30{,}35[}$.

Median: rank 105 lies in $[30, 35[$:

$$Me = 30 + 5\frac{105 - 80}{60} = \boxed{32.08}.$$

Quartiles:

$$Q_1 = 26 + 4\frac{52.5 - 48}{32} = \boxed{26.56},$$

$$Q_3 = 35 + 5\frac{157.5 - 140}{35} = \boxed{37.50}.$$

Using midpoints,

$$\sum n_i c_i = 6665, \qquad \bar{x} = 6665/210 = \boxed{31.74 \text{ euros}}.$$

The modal class has the greatest density; the median divides the observations into two equal halves; the mean is the average based on class midpoints; Q_1 and Q_3 delimit the central 50% of the distribution.

Detailed solution – Exercise 3

If 60% can pay, 40% must be below the price. The target cumulative count is

$$0.40(210) = 84.$$

Since the cumulative count is 80 at 30 euros and 140 at 35 euros, interpolate in $[30, 35[$:

$$p = 30 + 5\frac{84 - 80}{60} = \boxed{30.33 \text{ euros}}.$$